

WEEK2 GATE2

1. Executive Summary

API automation testing was performed using REST Assured, JUnit 5, JDBC, JSON Schema Validation, and Testcontainers.

2. Scope

APIs Covered

- Authentication APIs
- Product Catalog APIs
- Cart APIs
- Order APIs
- Order Lifecycle APIs

Validation Areas

- Functional Testing
 - Authentication Testing
 - Negative Testing
 - Schema Validation
 - Database Reconciliation
 - End-to-End Workflow Validation
-

3. Environment

Component	Details
Language	Java
Framework	JUnit 5
API Tool	REST Assured
Build Tool	Maven

Component	Details
Database	MySQL
Database Access	JDBC
Contract Validation	JSON Schema
Containerization	Testcontainers

4. Scenario Coverage

AuthenticationTest

Verify API security and access control.

1. Generate customer token.
 2. Access secured endpoints using valid token.
 3. Verify successful authentication.
 4. Access endpoints without token.
 5. Access endpoints with invalid token.
 6. Validate error responses.
-

ProductCatalogBaselineTest

Validate product catalog functionality.

1. Send GET request to retrieve product details.
 2. Verify status code is 200.
 3. Validate product ID.
 4. Validate product name.
 5. Validate product price.
 6. Validate response schema using Product.schema.json.
 7. Validate product list schema using ProductList.schema.json.
-

CreateOrderTest

Validate successful order creation and database persistence.

1. Verify database connectivity.
2. Generate authentication token.
3. Clear existing cart.
4. Add product to cart.
5. Verify cart update response.
6. Create order request.
7. Validate HTTP 201 Created response.
8. Validate response using Ops_OrderSchema (Partner.schema.json).
9. Verify order status is CREATED.
10. Extract generated Order ID.
11. Query MySQL database.
12. Verify order record exists.
13. Verify Order Number matches API response.
14. Verify Order Status matches API response.
15. Delete created test data after execution.

FlowTest (Ops_OrderLifeCycle)

Validate complete order lifecycle workflow.

1. Authenticate customer.
2. Retrieve product catalog.
3. Select product.
4. Add product to cart.
5. Verify cart details.
6. Create order.
7. Validate order creation response.
8. Validate response against Ops_OrderLifeCycle schema.

9. Verify order status transitions.
10. Validate business workflow completion.
11. Verify order details in database.

NegativeAssertionsTest

Verify API behavior for invalid scenarios.

1. Send requests with missing mandatory fields.
 2. Send requests with invalid data.
 3. Access secured APIs without token.
 4. Access secured APIs with invalid token.
 5. Access invalid resources.
 6. Validate HTTP error codes.
 7. Validate error messages.
 8. Verify API stability under invalid requests.
-

5. Schema Validation

JSON Schemas Validated

- Product.schema.json
- ProductList.schema.json
- Ops_OrderSchema (Partner.schema.json)
- Ops_OrderLifeCycle Schema

Validation Checks

- Required fields
- Data types
- Object structure
- Array structure
- Property constraints

XML Validation

- product.xsd

Result: All API responses successfully matched their contract definitions.

6. Database Reconciliation

Validation Process

1. Create order through API.
2. Capture generated Order ID.
3. Retrieve corresponding record from MySQL database.
4. Compare API response with database values.

Fields Verified

- Order ID
- Order Number
- Order Status
- Record Existence

Cleanup

Created test orders are automatically deleted after execution to maintain database consistency.

Result: Database records matched API responses successfully.

7. Execution Results

Test Class	Result
AuthenticationTest	Passed
ProductCatalogBaselineTest	Passed
CreateOrderTest	Passed
FlowTest (Ops_OrderLifeCycle)	Passed
NegativeAssertionsTest	Passed

Overall Result

- Functional Scenarios: Passed
- Authentication Scenarios: Passed
- Negative Scenarios: Passed
- Schema Validation: Passed
- Database Validation: Passed
- End-to-End Workflow Validation: Passed